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Unveiling the path towards sustainability: is there a research interest on sustainable goals?

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Introduction

Society in the 20th century has faced the effects of an unsustainable production model that led to global awareness of this topic. Different summits and conferences were held in which sustainability and sustainable development (SD) were the core discussion: this fact can be interpreted as a growing compromise from countries to work together on that issue and a topic that has become as policy priority setting. One of the most important was the Millennium Summit celebrated in 2000 which led to the establishment of the 8 Millennium Development Goals (MDGs). These goals were criticised for not be adequately aligned with human rights standards and principles and were relevant for poor countries only (Fukuda-Parr, 2016), and rich countries were supposed to add solidarity and assistance with finances and technology (Sachs, 2010). As a result, developing countries were made substantial progress towards the achievements of the goals. A framework with 17 Sustainable Development Goals (SDGs) indicators was established later with the Agenda 2030 (Rio, 2012). These indicators are an improved adaptation of the MDGs, and as Griggs stated (2013), they gave more attention to issues of environmental sustainability than the previous one. As well, these goals have a more ambitious vision because it assumes every country should assume responsibility and goes beyond international cooperation, but also focusing on eliminating discrimination and inequalities within the countries (Leal Filho et al., 2019).

Sources and methodology

The main objective of this paper is to analyse the scientific research on this subject and propose a methodology for classifying SDGs. Figure 1 illustrates the research methodology of this study.

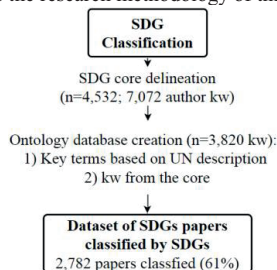


Figure 1. Research methodology

Results

4,532 documents were detected from 2000 to 2017 and this can be defined as the core of SDGs research. From this, 3,328 documents were from Millennium Development Goals (MDG) and 1,426 from Sustainable Development Goals (SDG) with an overlap of 218 documents between both. The cumulative average growth rate (CAGR) of scientific production during the period is 42.36% and, this growth, can be explained by the significant penetration of this topic in the Academia and research. During the period of study, the major increase of the scientific production can be related with the last years (2014-2017) with an increase of 110.39%, which coincides with the official launch of SDG in 2015.

By checking the co-occurrences map based on keywords, on the first years (2000-2005), it is observed that topics were related with health and well-being with terms such as malaria or mortality, but also with poverty or education. Besides, only keywords related to 'developing countries' or 'Africa' appeared, denoting the scope of these goals were focused on developing countries. In a more recent co-occurrence map (2012-2017), a more variety of topics was identified. Special emphasis should be taken to a node that includes terms such as policy or politics, which denote the interest to involve policymakers' decisions.

An ontology was created based on central keywords in the SDG United Nations description (e.g. 'sanitation' was classified into 'SDG6'). From the core research output, 7,072 author keywords were identified in the period and also were targeted into the goals. Only keywords with a clear focus on the SDGs were preliminary classified (e.g. 'Malaria' was classified into 'SDG3 Good Health and Well-being'). 3,820 keywords constituted the ontology and 2,782 papers (61%) from the core were classified. One paper could be multi-assigned. Then a co-occurrence network based on document relations on the SDGs was created (Figure 2). This shows the relations between SDGs papers and is divided into four clusters: Cluster no. 1 is related with urban settlement, its components and dynamics: sanitation (SDG6), responsible consumption and production (SDG12) and also with water from oceans, seas or marine resources (SDG14). Cluster no. 2 is related to governance and partnership for achieving SDGs and has the strongest edges between

the nodes, denoting a strong relation of these SDGs to have the topic of this goals published together. Besides, it has a strong connection with health node (SDG3). Cluster no. 3 is more environmental-related (climate-change aspects, land or energies); as well, it is linked with 'Industry, innovation and infrastructures' (SDG9) showing their link with the effects of the industry on the environment, with impact effects. Cluster no. 4 is related with socio-economic aspects (poverty, education) and labour aspects (work) and to reduce inequalities (SDG10).

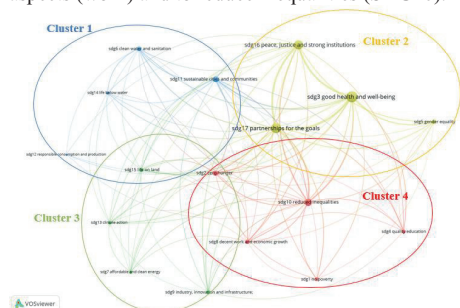


Figure 2. Co-occurrence map of SDG classification in scientific output (2000-2017).

Discussion and conclusions

Achieving long-term sustainability has become a challenge for all countries. As Salvia (2019) pointed out, the success on SDGs will rely on the strengthened collaboration of its actors. The involvement of the countries in the sustainable goals and their inclusion in global policy debates and national policy planning denotes the interest of global leaders in to achieve a more sustainable growth.

Is there a SDGs research?

This growth continues the findings of the research by Hassan et al. (2014) regarding SD scientific output, demonstrating the recent interest on this topic by the research community. It should be remarked the growth on the last 4-year period (increase of 100.39%) which coincides with the launched of the SDGs.

What topics are addressed in SDG scientific core production?

From the keyword co-occurrence map, it is stated that MDGs were more developing countries focused and did not consider environmental issues in comparison with SDGs. Topics were more related to health (SDG3), poverty (SDG1) and education (SDG4). With SDGs, the main difference is these ones implies the involvement of all countries with 'no one left behind' lemma. A new node include policy, denoting the involvement of policymakers. As well, with the growth and expansion of urbanization worldwide (Ramírez et al., 2016), cities

and communities (SDG11) has appeared as a node on scientific literature.

What are the SDGs relations between papers in the core scientific literature?

An ontology classification approach by SDGs is presented in order to determine which ones are the most focused in the scientific literature and relations between them. Partnership (SDG17) and the strong institutions (SDG16) are the most predominant, showing a strong connections between these goals. This goes in line with Waage et al. (2015), which stated the SDGs will mainly act in governance and partnership among the subscribing states. As well, a cluster of environmental-related can be identified, with SDG related to environment and its interactions (SDG13- Climate action, SDG15- Life on land).

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